Review - The Exponential Model

So far we've considered growth models that look like:

$$\frac{dP}{dt} = kP,$$

for some $k > 0$. This means that the *relative growth rate* is constant:

$$\text{relative growth rate} = \frac{1}{P}\frac{dP}{dt} = k.$$

Can you think of situations where this is not a good model for growth?

Limited Resources - The Logistic Model

- When resources are limited, we expect that that relative growth rate gets larger/smaller (cross out the wrong answer) as population grows. Why?

- The simplest decreasing function of P is a _____ function. This can be written as _____.
- So we get $\frac{1}{P}\frac{dP}{dt} =$ _____.
- For later convenience, we rearrange this as follows:

In general, when there is a limit L to how large $P(t)$ can get, a good model might be:

$$\text{relative growth rate} = \frac{1}{P}\frac{dP}{dt} = k\left(1 - \frac{P}{L}\right).$$

- When $P \approx 0$, then relative growth rate is approximately _____
- When $P \approx L$, then relative growth rate is approximately _____

This is called the **logistic growth model**.

1. Are there any equilibrium solutions?

2. Draw a slopefield for $\frac{dP}{dt}$ and classify any equilibrium solutions as stable or unstable. Sketch two solution curves: one with $P(0)$ near 0 and one with $P(0) > L$.

3. Draw a graph of $\frac{dP}{dt}$ as a function of P. For what value of P does $\frac{dP}{dt}$ reach its peak?

4. Assuming $P(0) = P_0$, check that the function $P(t) = \frac{L}{1+Ae^{-kt}}$ solves the initial value problem, and find A in terms of L and P_0.

Summary:

The solution to the logistic differential equation:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$

with initial condition $P(0) = P_0$ is the logistic function:

$$P(t) = \frac{L}{1 + Ae^{-kt}} \qquad \text{with} \qquad A = \frac{L - P_0}{P_0}.$$

Also:

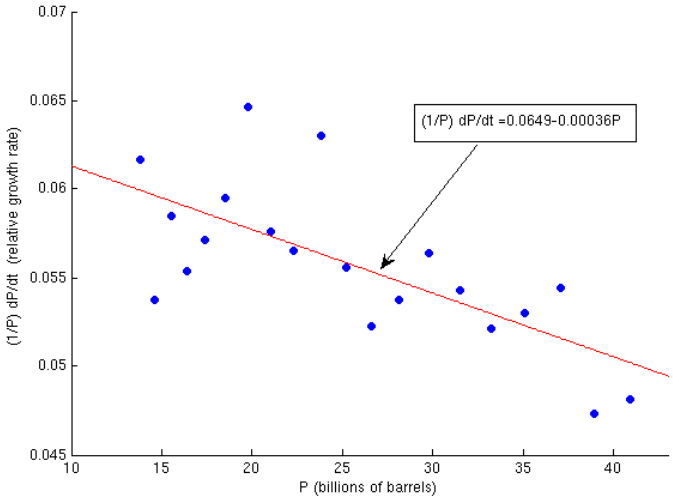
For a logistic function, the maximum value of $\frac{dP}{dt}$ occurs when $P = \frac{L}{2}$

US Oil Production

Let $P(t)$ be the **total** amount of oil (in billions of barrels) produced in the US since 1859, the year that oil production began. Assume that $P(t)$ grows according to the logistic model $\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$.

1. What is the meaning of L in this context?

2. The data below are based on estimates for both P and $\frac{dP}{dt}$ for the years 1931-1950. A best fit line is shown with its equation. Use it to estimate k and L. (Hint: $\frac{1}{P}\frac{dP}{dt} = k\left(1 - \frac{P}{L}\right)$. The expression on the right of that equation is the best fit line...)



3. If we let 1950 be time $t = 0$, then $P(0) = 40.9$ billion barrels. Find a formula for $P(t)$. You may use the solution to the logistic equation given at the top of the previous page.

4. According to this model, what year will oil production peak in the US and what will be the peak oil production amount? (Hint: if total oil produced is $P(t)$, what is oil production per year? Now use question 3 on page 2.)

(Note: US oil production actually peaked at 3.5 billion barrels per year in 1970)

Running

Jane is really out of shape and currently runs at a 3 mi per hour pace. She begins training on Jan. 1, 2013; her pace improves dramatically, although she will never be able to run faster than 7 mi per hour. Let $v(t)$ be Jane's pace (in mi per hour) t days after Jan. 1.

1. Assuming that her pace $v(t)$ grows according to a logistic model, write down an IVP for $v(t)$.

2. If $v(100) = 5$, find an explicit formula for $v(t)$, then plot it. (Hint: you will have to calculate k...)